Hardest Materials

An exploration into three tools and their hardness properties

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Scissors

Sample Description

The sample in question is the *TWIN L Kitchen Shears* made by the company *Zwilling*. *Zwilling* is a German company that was founded in 1731, making it one of the oldest and also largest manufacturers of kitchen knives for domestic and professional use in the world. The knife used for the experiment was bought on 京东 for ¥100 and it is made in China.

According the the manufacturer, the scissors can be used for snipping herbs, cutting foods or safely opening packages. The scissors weight 0.10kg, they are 20.50cm long, with the blade having a length of 9.50cm and the total height of the scissor is 1.30cm. *Zwilling* claims that the scissors have a relatively low weight, while still offering very precise cutting abilities. The scissors are made out of two material groups; there is a polymer (plastic) grip, along with another polymer connection point holding the two blades together. And then there is the blade itself, which is made out of stainless steel. In the next section, we will have a more detailed look at the material used for the blade of the scissors, as this will be the material which we will test for its hardness.

Material Analysis

According to the manufacturer, the scissors are made out of "stainless special formula steel for durability", meaning that a specially engineered type of stainless steel is used that allows for a higher durability of the material. The blade can be sharpened, which indicates that there is a certain requirement of ductility so that the blade will not shatter during the sharpening process. Furthermore, the scissors are dishwasher-safe, indicating that they blade as well as the polymer structure can withstand relatively high water temperatures. The blades are made by the manufacturing method of stamping, which means using a high-pressure press to cut and shape flat sheets of metal into the desired blade form. This process involves placing a sheet of metal, typically steel, between a die and a punch. The press then forces the punch into the die, cutting the metal to the precise shape and size required for the scissor blades. What is important to note however, is that usually after obtaining the scissor blades through stamping, the blades normally undergo further processing such as heat treatment for hardness or grinding for sharpness.

Regarding the hardness of the material, the manufacturer actually claims that the scissors have a hardness ranging from 50-54 HRC. In the material testing done below, we will verify this specification.

Material Preperation

Before we could start observing the microstructure of the material and test its hardness, we had to first prepare the material so that it would be in an appropriate state for testing. What that meant was that we first had to take a small sample of the scissor blade, which the TA kindly offered to do, and he put the sample in a rubber block, so it was easier for us the perform the later processing. After cutting the material, we had to sand the sample with continuously finer sand paper and then we polished the surface of the material. After that, the last step before observation was to etch the material using 4% nitric acid and alcohol. After the first etching, the result seen in the microscope was far from ideal, so I actually had to etch the sample multiple times before I obtained an acceptable image on the microscope. This final microscopic image of the sample is discussed in the next section.

Microstructure

The picture obtained from the microscope is as below:

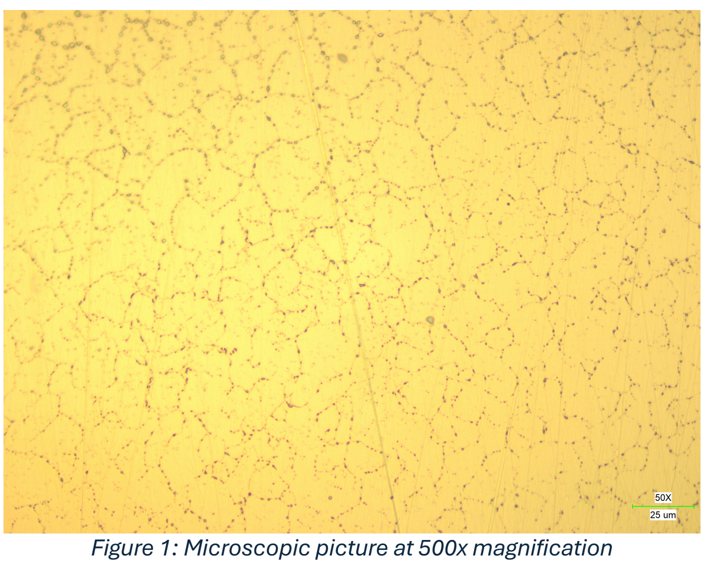
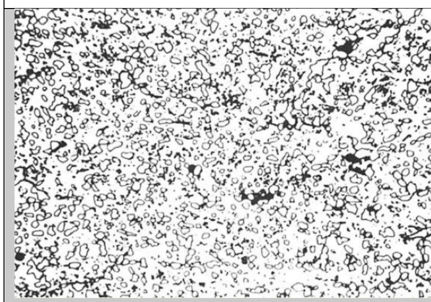
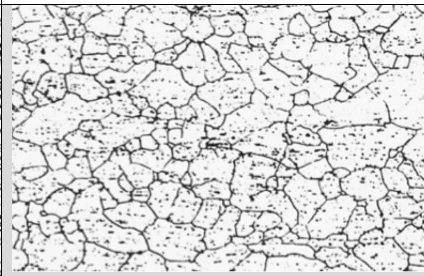
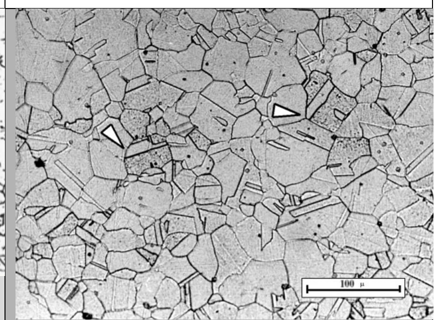


Figure 1: Microscopic picture at 500x magnification

Now, to analyse the microstructure, we can consider what we already know from the product description; which is that the scissor blades are made out of stainless steel. In class, we have discussed three types of stainless steels, whose microstructures are presented below:

Martensitic	Ferritic	Austenitic
		

Just by comparing the pictures, we can see that only ferritic or austenitic materials seem plausible. The microstructure of ferritic stainless steel consists of a ferritic matrix and dispersed carbide particles, while the austenitic stainless steel has an austenitic matrix with dispersed carbide particles. To test if the sample is ferritic or austenitic, you could look at the magnetic properties of the sample, because ferritic stainless steel is magnetic whereas austenitic stainless steel is not. Unfortunately, I did not perform this test, so to determine which type of stainless steel we have present in the sample, it is helpful to look at the material properties of ferritic and austenitic stainless steels. Ferritic stainless steels offer moderate corrosion resistance, good strength and good ductility but are brittle and don't have very high toughness. Austenitic stainless steels on the other hand offer moderate strength, high formability and they have a better corrosion resistance than other steels. When you are working with food, which the knife we are analysing is made for, then it is important to try to avoid corrosion on materials that have direct contact with food, to not contaminate the food. From this perspective, it seems more likely that the sample is made out of austenitic stainless steel, as it offers high corrosion resistance.

The microscopic picture therefore shows the austenitic matrix and the numerous little dots on the picture represent the carbide particles dispersed in the matrix.

Hardness Measurement

After preparing the material and observing its microstructure, we further used a material hardness tester device to test the hardness of the material. We used the Vickers hardness test, which gave us a measure of hardness in units of Vickers Pyramid Number (HV). To eliminate small inaccuracies in the measurements, we took three measurements and then calculated the average value, which in my case resulted in a hardness of 674 HV. Using a conversion table, we can convert the Vickers Hardness Number (HV) into a Rockwell C Scale Hardness Number (HRC), which will give us a value of 55 HRC. Comparing this to the specifications given by the manufacturer, we can see that this value is even slightly above the range specified, meaning that the scissors fulfilled the manufacturing requirements. In general, one can say that this is a mediocre hardness but it is enough for the use case the knife is designed for.

Reasons for Hardness

The main alloying element for stainless steel is chromium, of which a percentage of at least 11%wt is required. The primary determinant of hardness in austenitic stainless steel material is the ferrite content. That is why ferritic stainless steels are usually harder than austenitic stainless steels. The amount of ferrite which is present in a given casting depends primarily on the composition, and secondarily on the section thickness and the heat treatment the part received. Thicker sections generally have higher ferrite contents than thinner sections. The heat treatment effect is due to the temperature used; the precise effect is often not predictable since it also depends on the composition. Elements such as chromium, molybdenum, silicon, and columbium (niobium) can promote the formation of ferrite, making the material even harder. However, there are also elements which can promote the formation of austenite, which hinders the formation of ferrite. This elements include nickel, manganese, carbon, nitrogen, etc. For austenitic steel, you will get a rounded combination of moderate strength and good corrosion resistance. If you wanted higher strength, you would have to increase the amount of ferrite, which in turn decreases the corrosion resistance ability of the material.Therefore, it is the manufacturers goal to find a composition that strikes a good balance and is suitable for a kitchen knife.

Material Processing Technique

As mentioned before, the scissors are made by stamping them out of a sheet of metal. However, after stamping, heat treatment is required to ensure the scissor blades have the desired material properties. Usually, scissors are subjected to cold-hardening after stamping to improve their stain resistance. This is done by a process called cryogenic tempering, which involves immersing the finished knife blades in liquid nitrogen. This process is required to get full hardness from most stainless knife steels, as it promotes the conversion of austenite to martensite. According to the requirements for the scissors, this conversion needs to be controlled to form just the right amount of martensite, austenite and ferrite which is required for the task the material is supposed to fulfil. For the scissors investigated, it is important to leave some austenite in order to obtain better ductility and corrosion resistance.

Wrench

Sample Description

The wrench is identified as a "多功能扳手活口活动扳手" which translates to "multifunctional spanner wrench with an adjustable jaw" and is a product of the "德力西" (Delixi) brand. It is designed with a large opening, making it suitable for use with various sizes of fasteners and is particularly highlighted as being suitable for use with bathroom fixtures and faucets. The total length of the wrench is 200mm. The wrench has an adjustable opening range that can accommodate sizes from 0 to 30mm. The weight of the wrench is 295 grams, indicating it is relatively lightweight for its size and intended use. The product was bought on Taobao at 26.90￥ and was made in China.

Figure 2 Chromium Vanadium Steel Wrench

Material Analysis

According to the seller, the wrench in question is crafted from a high-quality alloy known as chromium vanadium steel, which is renowned for its superior mechanical properties and durability in tool applications. Chromium vanadium steel is an engineered alloy that incorporates chromium and vanadium into its composition. The addition of these elements enhances the steel's hardness, wear resistance, and tensile strength. This type of steel is specifically chosen for applications where high durability and resistance to deformation under load are required.

The wrench is forged from this alloy, a process that involves shaping the metal while it is still in a malleable state. Forging imparts a uniform grain structure to the steel, which contributes to the overall strength and integrity of the wrench. The manufacturing method also includes a series of heat treatments that further enhance the material properties. The wrench then undergoes a series of heat treatments, which are critical in determining the final hardness and mechanical properties of the tool. These treatments may include quenching and tempering processes that harden the steel and relieve internal stresses, respectively.

The material's ductility is also an important factor, allowing the wrench to be sharpened and maintained without the risk of shattering or breaking. This property is essential for tools that require frequent sharpening to maintain optimal performance.

The manufacturer specifies that the wrench is designed to achieve a hardness level greater than 60 HRC (Rockwell C scale). This high level of hardness ensures that the wrench can withstand significant torque and resist wear during use.

The manufacturer also claims that the wrench features an electroplated black nickel coating, which serves as a protective barrier against rust and oxidation. This advanced surface treatment not only enhances the wrench's durability by reducing the risk of corrosion but also contributes to its aesthetic appeal with a sleek, professional finish. The black nickel coating is particularly beneficial in environments where tools are exposed to moisture, chemicals, or other corrosive elements, ensuring that the wrench maintains its integrity and functionality over time. Nevertheless, the claims will have to be verified through future experiments.

Specimen Preparation

Prior to examining the material's microstructure and measuring its hardness, it is essential to prepare the sample to ensure it was suitable for testing. The process began with cutting a small segment of the scissor blade. The teaching assistant graciously offered to do it for us. This sample was then embedded in a glossy transparent elastomer to facilitate subsequent procedures.

The initial step post-cutting was to refine the sample's surface using a sequence of progressively finer sandpapers. This was followed by a meticulous polishing process to achieve a smooth finish. The penultimate stage in preparation involved etching the sample, which was accomplished using a solution of 4% nitric acid mixed with alcohol. The initial etching did not yield the desired clarity under the microscope, necessitating multiple rounds of etching to achieve a satisfactory result. The refined microscopic image obtained after these iterations will be the subject of discussion in the subsequent section.

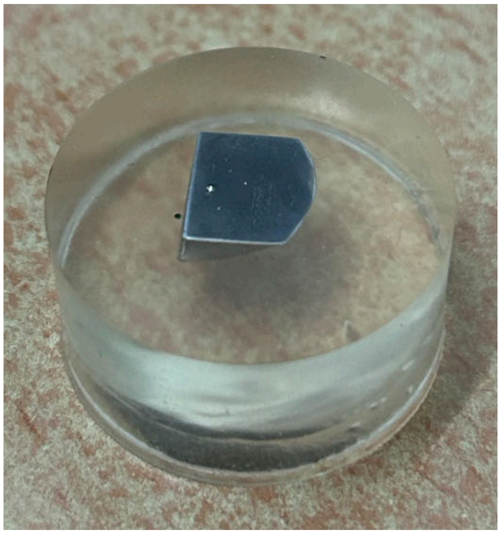


Figure 3 The Specimen

Microstructure of Specimen

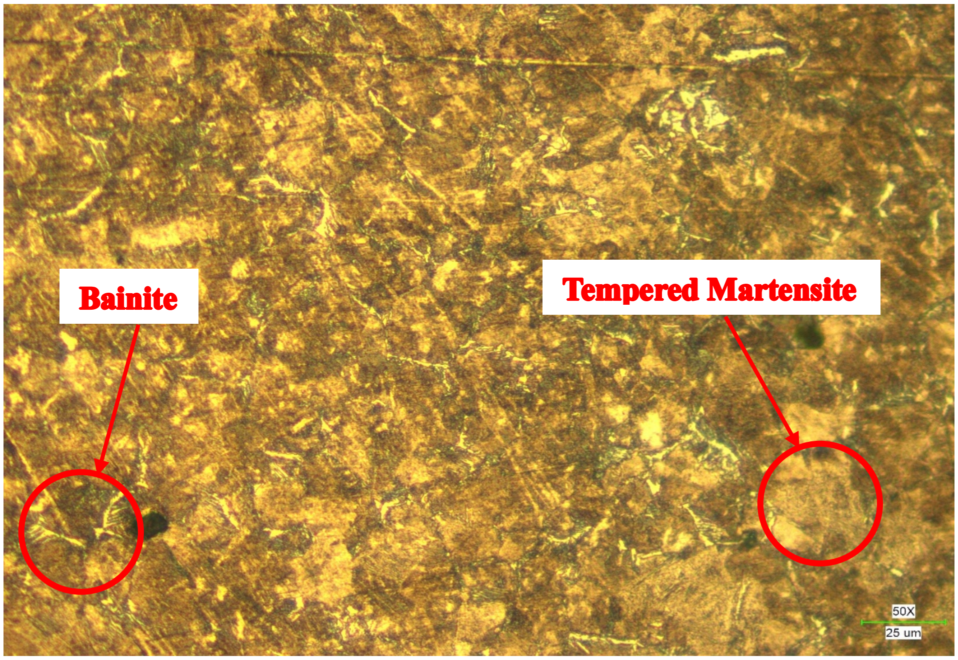


Figure 4 Microstructure of Specimen

The micrograph shows a pattern of light and dark regions, suggesting a dual-phase structure. The structure appears somewhat grainy, indicating a mixture of different phases. The microstructure of specimen consists of tempered martensite, which is typical for heat-treated high-strength steels like Cr-V steel. The light regions could be tempered martensite, and the darker areas might represent retained carbides or secondary phases. It also consists of bainitic structure. There might also be some retained austenite.

As for mechanical properties of specimen based on microstructures, tempered martensite offers high hardness, high tensile strength, yield strength (critical for high-load applications) and improved ductility compared to untampered martensite due to reduced internal stresses. The presence of carbides or secondary phases further increases hardness. Next, bainite provides moderate hardness, higher than ferrite but lower than martensite, contributing to wear resistance. Overall, the microstructure suggests that the material has a high hardness level, reasonable ductility, balancing toughness and resistance to crack propagation, beneficial for tools and wear-resistant applications.

The possible heat treatments that Chromium vanadium steel undergoes to optimize its mechanical properties are quenching, tempering and annealing.

Hardness Measurement

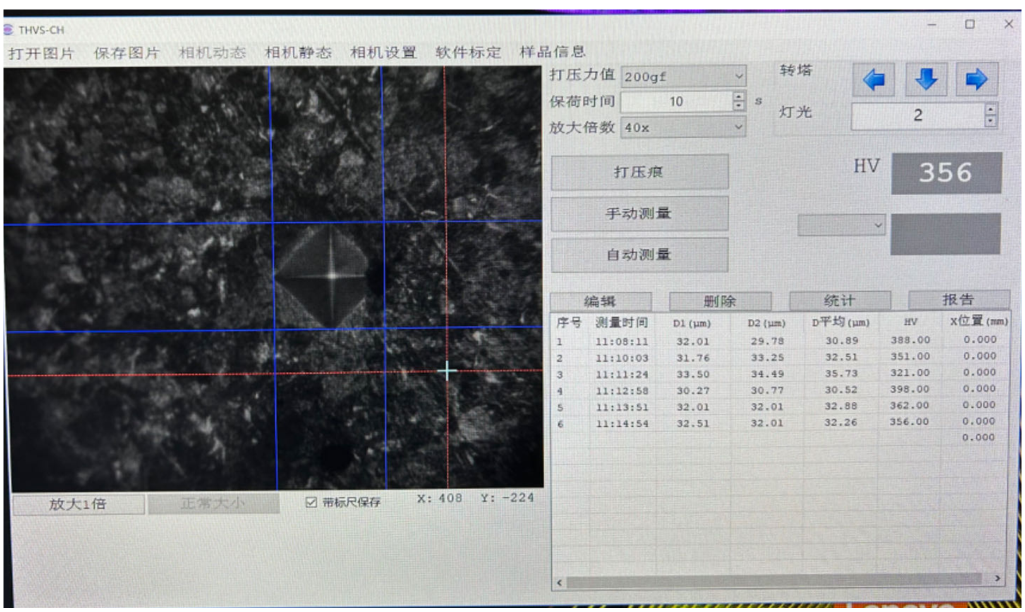


Figure 5 Vickers Hardness Test Result

Figure 5 shows the indentation made by the Vickers diamond indenter on the steel sample. The indentation is a clear, well-defined diamond shape, indicating a successful test. The displayed hardness value is 356 HV (Vickers Hardness). This value is derived from the indentation's diagonals measured on the image, using the Vickers formula. The test is performed with the following test parameters:

- Applied Force: 200 gf (gram-force), which is a standard load for microhardness testing.
- Holding Time: 10 seconds, ensuring accurate and consistent indentation.
- The test and observation were conducted at 40 times magnification, providing detailed visibility of the indentation.

The table in Figure 5 shows multiple measurements of the indentation diagonals, indicating the precision and repeatability of the test. The Vickers hardness value of 356 HV suggests that the material has high hardness, consistent with the presence of tempered martensite or bainite observed in the microstructure. This level of hardness is typical for heat-treated chromium vanadium steels, which are known for their excellent wear resistance and ability to maintain high hardness after heat treatment.

Alloyed Hammer

The third product purchased and tested was a hammer and is pictured in Figure 6. The manufacturer cites the hammer composition as a 'Cemented Tungsten Carbide-Cobalt Alloy'[1].

Material Composition

Figure 6: Cemented WC-Co Alloy Hammer

From lecture content and textbook and other research, typically consists of two main phases:

1. **Tungsten Carbide (WC) particles**. These are the hard, primary phase that provides the exceptional hardness and wear resistance. Tungsten carbide is a ceramic material, however, similar to other advanced ceramics, tungsten and carbon bonds are primarily covalent (not ionic), and share ceramic properties such as high hardness, high melting point, and brittleness.

2. **Cobalt Binder**: The cobalt phase acts as a metallic binder, holding the WC particles together. The Co binder is relatively softer and fills in the space between the WC particles. As cobalt is unalloyed in this configuration, cobalt atoms are bonded metallically, providing metallic properties such as ductility, toughness, and electrical / thermal conductivity.

Considering that WC-Co is composed of both a ceramic and a metal, the hammer's composition can thus be classified as a composite. Specifically, it can be classed as a *metal-matrix composite*, whereby the composite material is composed of a metal or alloy matrix embedded with reinforcing particles, fibers, or whiskers, typically made of ceramics or other high-strength materials.

Methodology

Grinding

To assess the material's hardness, a part of the hammer was extracted and mounted in resin by a skilled metallurgical specialist. The mounted specimen was then coarsely grinded for approximately one hour to remove visible chains on the specimen's surface while using the

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[1] '钨钢锤单晶硅用碳化钨破碎锤多晶硅榔头非标订制钨钴合金钨钢锤子'Product Description.

coarsest (180 grit size) abrasive disc. Given the effort and time committed to this endeavor, the laboratory technician advised to assess what was grinded, albeit *seemingly* incomplete.

Etching

Following this arduous stage, 4% nitric acid and alcohol was applied to the specimen for 10 seconds. After etching, the material was rinsed and heat-air dried.

Microscope

To assess the material's microstructure, the specimen was placed beneath an electronic microscope and two images were captured. One image was captured at 4x magnification, and the second image was captured at 50x magnification.

Hardness tester

To assess the material's hardness, Vickers's hardness (HV) test was applied. The test's load was 500 grams-force and it was applied for 10 seconds, and the indentation was observed under a 40x magnification to measure the material's hardness. Three hardness values were derived from the specimen: 1,427, 1,469, and 1,447. The averaged HV is therefore 1,448.

Microscopic Analysis

Image 1

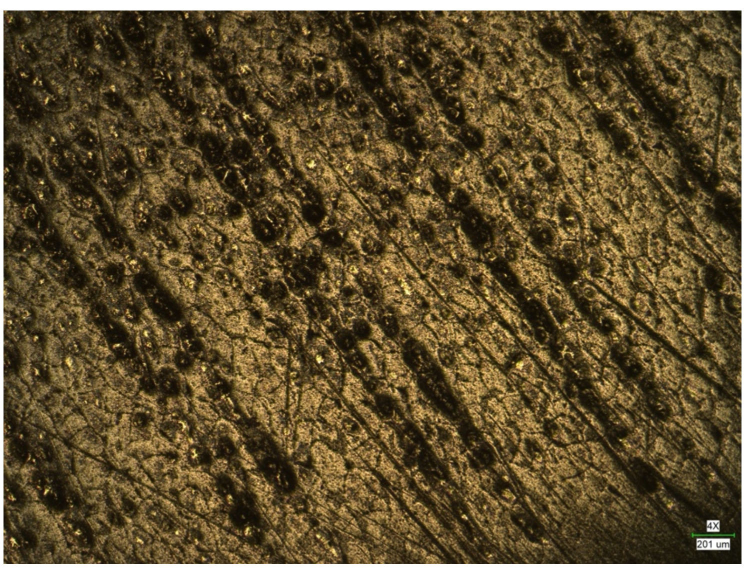

***Figure 7**: 4x magnified image of the WC-Co hammer sample.*

The dark spots and regions in **Figure 7** likely correspond to tungsten carbide particles. These particles are dispersed throughout the structure, providing the bulk of the material's hardness. WC particles are usually dark or black in microstructure images due to their high density and hardness. The lighter regions between the dark particles are likely the cobalt binder. This phase likely helps to absorb and distribute the stresses, adding toughness to the material.

Figure 8 provides a clearer view of the fine details within the tungsten-cobalt microstructure. The light-colored regions and network structure suggest the presence of the cobalt binder phase surrounding the WC particles. The relatively uniform distribution of WC particles within the Co matrix is also visible, indicating a well-sintered material. This uniformity is crucial for ensuring consistent mechanical properties throughout the material

Image 2

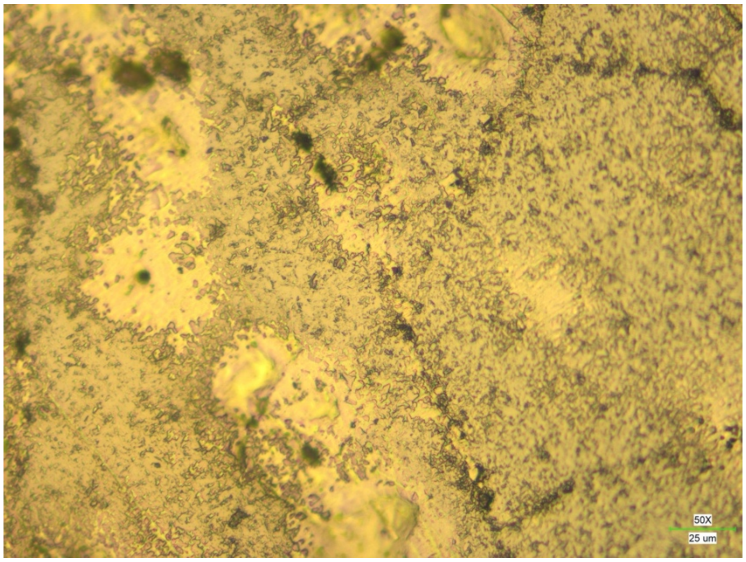

***Figure 8**: 50x magnified image of the WC-Co hammer sample.*

Hardness and functionality

As indicated above, the primary source of hardness in cemented carbides is the tungsten carbide particles. These particles are extremely hard, providing excellent wear resistance and contributing significantly to the overall hardness of the material. Reasons for this high hardness may relate with strong covalent bonding and the bond strength between the two as well as the hexagonal crystal structure, which is relatively more resistant to deformation (because of their fewer slip systems). Additionally, tungsten as a pure metal has exceptionally high hardness.

The cobalt binder phase, while softer than WC, plays a critical role in enhancing the toughness of the material. It allows the composite to withstand impacts and absorb energy, reducing the brittleness that pure WC might exhibit.

Functionally, and with a combination of both these elements, the exhibited properties are desirable for a high-quality hammer, considering the hammer needs high impact strength and toughness to meet the needs of its in-service conditions.

Conclusion

This report explores three tools and their respective hardness measurements. The report seeks to identify which tools have the highest hardness and the reasons behind their higher hardness. The three tools assessed in this report include scissors, a wrench, and a hammer .

All in all, we found that the hardest material of the three we inspected was the 'Cemented Tungsten Carbide-Cobalt Alloy' of the hammer with an HV value of 1448. The tungsten carbide particles found in the alloy are the main reason for the extreme hardness of this material. Considering the applications of the hammer, it should be no surprise that it would have the hardest material. On second place, we have the kitchen scissors with an HV value of 674. The scissors are designed to have a good combination of hardness, corrosion-resistance and ductility; therefore, the hardness is not as high as materials designed with high hardness requirements in mind. The main microstructure determining the hardness of the scissor blades is the amount of ferrite present, but as austenite is also needed for better ductility, the hardness is slightly decreased. On the last place we find the wrench, which was found to have an HV value of 356. It gets its hardness from the tempered martensite and carbide particles present but to also offers good ductility due to reduced internal stresses.

In summary, we saw that the hardness of steels in daily life varies strongly according to their intended application. Whereas in some applications, a very high hardness value is necessary, other applications might call for more rounded material property, striking a good balance between hardness and ductility. Therefore, when researching materials properties, it is always important to keep their specific application in mind and interpret your results accordingly.